Correlation of Signals

Outlines

- Basics of Correlation of Signals
- Types of Correlation of Signals
- Comparison of Correlation and Convolution of Signals
- Relationship between Correlation and Convolution of Signals
- Applications of Correlation of Signals

Basics of Correlation of Signals

- Correlation measures similarities between two signals.
- If the Correlation between two different signals is zero, then there are no similarities between the two signals.
- If we have two signals $x_1(t)$ and $x_2(t)$, then the Correlation between two signals is defined as:

$$\Rightarrow x_{12}(\tau) = \int_{-\infty}^{\infty} x_1(t)x_2(t-\tau)dt$$

- If we have two Complex signals $x_1(t)$ and $x_2(t)$, then the Correlation between two signals is defined as:

$$\Rightarrow x_{12}(\tau) = \int_{-\infty}^{\infty} x_1(t){x_2}^*(t-\tau)dt$$

- Here, τ is delay parameter.

Types of Correlation of Signals

- ❑ There are Three Types of Correlation:
- ❑ Auto Correlation: It explains the Similarities of the Signal with its time-shifted version.

$$\Rightarrow \mathbf{x_{11}(\tau)} = \int_{-\infty}^{\infty} \mathbf{x_1(t)x_1(t-\tau)dt}$$

  - It is used to detect Periodicity
  - It is used for Noise Reduction
  - It is used to determine the Impulse response of the system
  - It is also used to determine Doppler shift in echo with RADAR.
- ❑ Cross Correlation: It explains the Similarities between the two different signals.

$$\Rightarrow \mathbf{x_{12}(\tau)} = \int_{-\infty}^{\infty} \mathbf{x_1(t)x_2(t-\tau)dt}$$

  - It is used to match Patterns between two signals
  - It is used to measure time delay between two signals
  - It is used to identify Peak Match and Time Shift Match between two signals.

Types of Correlation of Signals

- **Circular Correlation:** It explains the Similarities between the two different periodic signals.

$$\Longrightarrow \mathbf{x_{12}} = \mathbf{F^{-1}}[\mathbf{x_1}^*(\mathbf{k})\mathbf{x_2}(\mathbf{k})]$$

- $\mathbf{F^{-1}}$ is Inverse FFT.
    - It is used in OFDM for Synchronization
    - It allows fast computing due to FFT
    - It avoids the edge effect seen in Linear Correlation due to Wrapping around.

Comparison of Correlation and Convolution of Signals

Parameters	Correlation	Convolution
Function	$\Rightarrow x_{12}(\tau) = \int_{-\infty}^{\infty} x_1(t)x_2{}^*(t-\tau)dt$	$\Rightarrow x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau)d\tau$
Purpose	▪ Measures Similarity between two signals.	▪ Describes Systems response
Complex Conjugation	▪ Yes	▪ No
Order of Signals	▪ Matters	▪ Does Not Matter
Applications	▪ Pattern Recognition	▪ System Response

Relationship between Correlation and Convolution of Signals

❑ If we take the complex conjugate of one signal and flip it, then it becomes a correlation.

$$\Rightarrow [x_1{}^*(t) * x_2(t)](-T) = x_{12}(T)$$

Applications of Correlation

- RADAR and SONAR: Target identification by comparing two signals {Transmitted and Received Signals}.
- Medical Imaging: Image Pattern Recognition.
- Speech Processing: Noise Reduction.
- Communication System: Signals Synchronization.

Properties of Correlation

Properties of Cross Correlation

- **Property 1:** Cross Correlation exhibits conjugate Symmetry.

$$\Rightarrow R_{12}(\tau) = R_{21}^{*}(\tau)$$

  - Cross Correlation does not exhibit general commutative, $R_{12}(\tau) \neq R_{21}(\tau)$.

- **Property 2:** Cross Correlation exhibits Orthogonal Property, if $R_{12}(0) = 0$.

$$\Rightarrow x_{12}(0) = \int_{-\infty}^{\infty} x_1(t) x_2^{*}(t) dt$$

$$\Rightarrow x_{12}(0) = \frac{1}{T} \int_{-T/2}^{T/2} x_1(t) x_2^{*}(t) dt$$

  - Here, $x_1(t)$ and $x_2(t)$ are orthogonal to each other.

Properties of Auto Correlation

- **Property 1:** Auto Correlation exhibits conjugate Symmetry.

$$\Rightarrow R_{11}(\tau) = R_{11}{}^{*}(-\tau)$$

  - This means that the real part of $R_{11}(\tau)$ is an even function of τ and the imaginary part of $R_{11}(\tau)$ is an odd function of τ.

- **Property 2:** Auto Correlation for Energy Signal is Energy at $\tau = 0$.

$$\Rightarrow R_{11}(0) = E = \int_{-\infty}^{\infty} x_1(t)x_1{}^{*}(t)dt$$

- **Property 3:** Auto Correlation for Energy Signal is Maximum at $\tau = 0$.

$$\Rightarrow R_{11}(0) \geq R_{11}(\tau)$$

- **Property 4:** Auto Correlation is periodic with the same period (T) as periodic signal itself.

$$\Rightarrow R_{11}(\tau) = R_{11}(\tau \pm nT)$$

- **Property 4:** Fourier Transform of Auto Correlation for Energy/Power Signal is Energy/Power Spectral Density.

$$\Rightarrow R_{11}(\tau) \overset{F.T.}{\longleftrightarrow} \psi(\omega)/S(\omega)$$

Proof of Properties of Correlation

- **Property 1:** Auto Correlation exhibits conjugate Symmetry.

$$\Longrightarrow R_{11}(\tau) = R_{11}^*(-\tau)$$

- This means that the real part of $R_{11}(\tau)$ is an even function of τ and the imaginary part of $R_{11}(\tau)$ is an odd function of τ.

$$\rightsquigarrow R_{11}(\tau) = \int_{-\infty}^{\infty} x_1(t)\, x_1^*(t-\tau)\, dt$$

$$\rightsquigarrow R_{11}^*(-\tau) = \left[\int_{-\infty}^{\infty} x_1(t)\, x_1^*(t+\tau)\, dt\right]^*$$

$$= \int_{-\infty}^{\infty} x_1^*(t)\, x_1(t+\tau)\, dt$$

- $t+\tau = y \Rightarrow t = y-\tau \Rightarrow dt = dy$
- $t \in (-\infty, \infty) \Rightarrow y \in (-\infty, \infty)$

$$= \int_{-\infty}^{\infty} x_1^*(y-\tau)\, x_1(y)\, dy = R_{11}(\tau)$$

❑ Property 2: Auto Correlation for Energy Signal is Energy at $\tau = 0$.

$$\Longrightarrow \mathbf{R_{11}(0) = E}$$

$$\rightsquigarrow R_{11}(\tau) = \int_{-\infty}^{\infty} x_1(t)\, x_1^*(t-\tau)\, dt$$

$$- \tau = 0$$

$$\Rightarrow R_{11}(0) = \int_{-\infty}^{\infty} x_1(t)\, x_1^*(t)\, dt$$

$$= \int_{-\infty}^{\infty} |x_1(t)|^2\, dt$$

$$= E$$

❑ **Property 3:** **Auto Correlation for Energy Signal is Maximum at $\tau = 0$.**

$$\Rightarrow R_{11}(0) \geq R_{11}(\tau)$$

$\Rightarrow (x(t) - x(t-\tau))^2 \geq 0$

$\Rightarrow x^2(t) + x^2(t-\tau) - 2x(t)x(t-\tau) \geq 0$

$\Rightarrow x^2(t) + x^2(t-\tau) \geq 2x(t)x(t-\tau)$

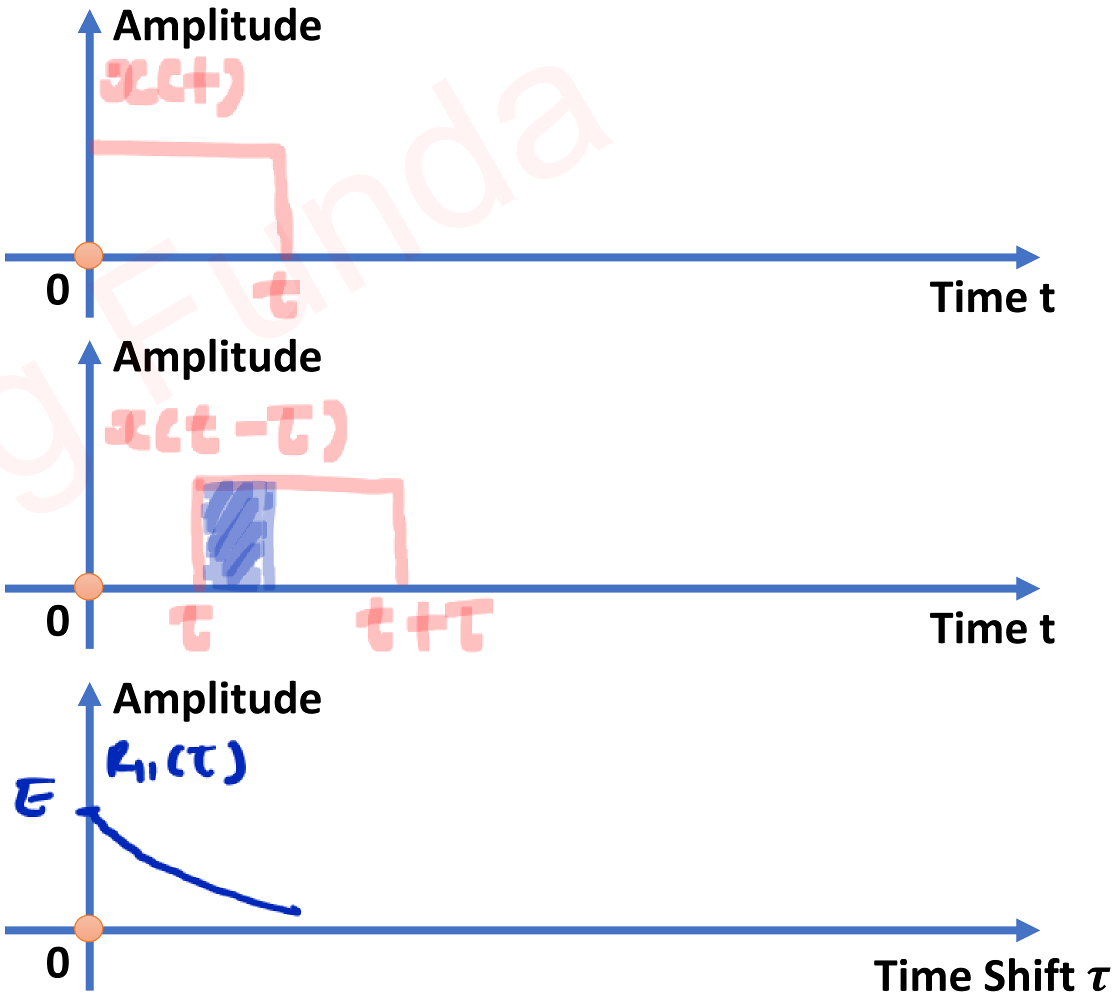
- take limits ($-\infty$ to ∞) w.r.t t.

$\Rightarrow \int_{-\infty}^{\infty} x^2(t)\,dt + \int_{-\infty}^{\infty} x^2(t-\tau)\,dt \geq 2\int_{-\infty}^{\infty} x(t)x(t-\tau)\,dt$

$\Rightarrow E + E \geq 2R_{11}(\tau)$

$\Rightarrow E \geq R_{11}(\tau)$

$\Rightarrow R_{11}(0) \geq R_{11}(\tau)$



Examples of Correlation

Example 1. Find the Cross Correlation of $x(n) = \{\underset{\uparrow}{1}, 1, 2, 2\}$ and $y(n) = \{\underset{\uparrow}{1}, 2, 3\}$.

$x(n)$ \ $y(-n)$	3	2	1
1	3	2	1
1	3	2	1
2	6	4	2
2	6	4	2

$\rightsquigarrow$ $\underline{R_{xy}(n)} \neq R_{yx}(n)$

$\rightsquigarrow$ $x(n) = \{\underset{\uparrow}{1}, 1, 2, 2\}$

$y(n) = \{\underset{\uparrow}{1}, 2, 3\}$

- $y(-n) = \{3, 2, \underset{\uparrow}{1}\}$
- $R_{xy}(n) = \{3, 5, \underset{\uparrow}{9}, 11, 6, 2\}$

Example 2. Find the Auto Correlation of $x(n) = \{\underset{\uparrow}{1}, 1, 2, 2\}$.

$x(n)$ \ $x(-n)$	2	2	1	1
1	2	2	1	1
1	2	2	1	1
2	4	4	2	2
2	4	4	2	2

- $x(n) = \{\underset{\uparrow}{1}, 1, 2, 2\}$

  $x(-n) = \{2, 2, 1, \underset{\uparrow}{1}\}$

- $R_{xx}(n) = \{2, 4, 7, \underset{\uparrow}{10}, 7, 4, 2\}$

Example 3. Find the Circular Correlation $R_{xy}(n)$ of $x(n) = \{\underset{\uparrow}{1}, 2, 3, 4\}$ and $y(n) = \{\underset{\uparrow}{1}, 1, 2, 2\}$.

$$R_{xy}(n) = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 2 & 1 & 1 & 2 \\ 2 & 2 & 1 & 1 \\ 1 & 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 17 \\ 15 \\ 13 \\ 15 \end{bmatrix}$$

$$= \{\underset{\uparrow}{17}, 15, 13, 15\}$$

Example of Auto Correlation

Example 1. Find the Auto Correlation of $x(t) = A\sin(\omega_0 t)$ and also find Power of given Signal.

$\hookrightarrow P = \frac{V_m^2}{2} = \frac{A^2}{2}$

$\rightarrow R_{xx}(\tau) = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} x(t)\,x(t-\tau)\,dt$

$\rightarrow x(t) = A\sin(\omega_0 t)$

$x(t-\tau) = A\sin(\omega_0(t-\tau))$

$\Rightarrow R_{xx}(\tau) = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} A^2 \sin\omega_0 t \,\sin(\omega_0(t-\tau))\,dt$

$= \lim_{T\to\infty} \frac{A^2}{2T}\int_{-T/2}^{T/2} 2\sin(\omega_0 t)\sin(\omega_0(t-\tau))\,dt$

$[\, 2\sin A \sin B = \cos(A-B) - \sin(A+B) \,]$

$$\Rightarrow R_{xx}(\tau) = \lim_{T\to\infty} \frac{A^2}{2T} \int_{-T/2}^{T/2} \cos(\omega_0\tau) - \underbrace{\sin(2\omega_0 t - \omega_0\tau)}_{=0}\, dt$$

$$= \lim_{T\to\infty} \frac{A^2}{2T} \cos(\omega_0\tau)\,[t]_{-T/2}^{T/2}$$

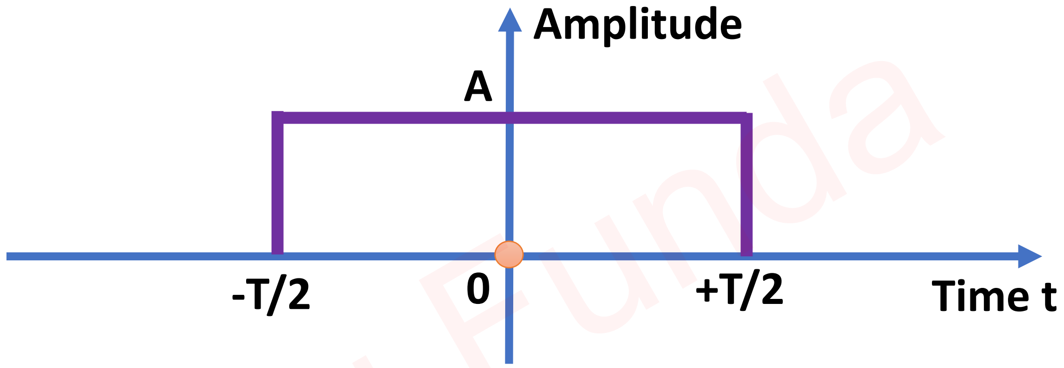
$$\Rightarrow R_{xx}(\tau) = \frac{A^2}{2} \cos(\omega_0\tau)$$

$\rightsquigarrow$ $R_{xx}(\tau)$ is periodic with same Freq./Period.

$\rightsquigarrow$ Power $= R_{xx}(\tau)\big|_{\tau=0} = \frac{A^2}{2}$

Example of Auto Correlation

Example 2. Signal $x(t)$ is as follows:

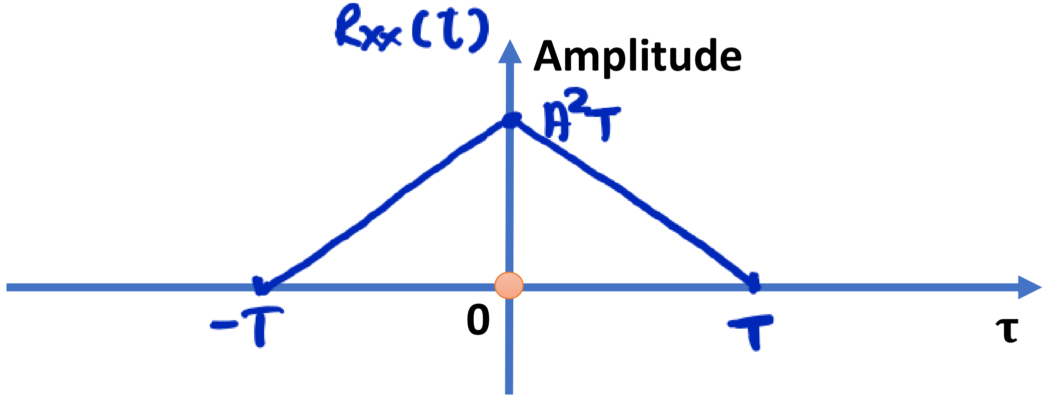


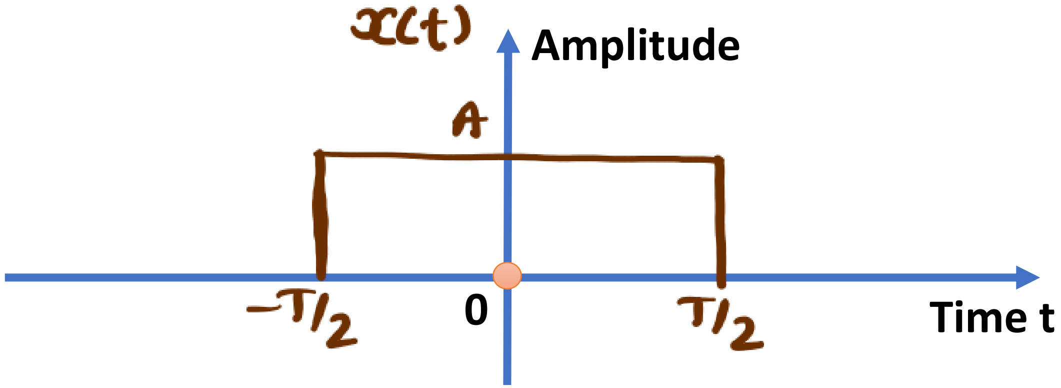
Find

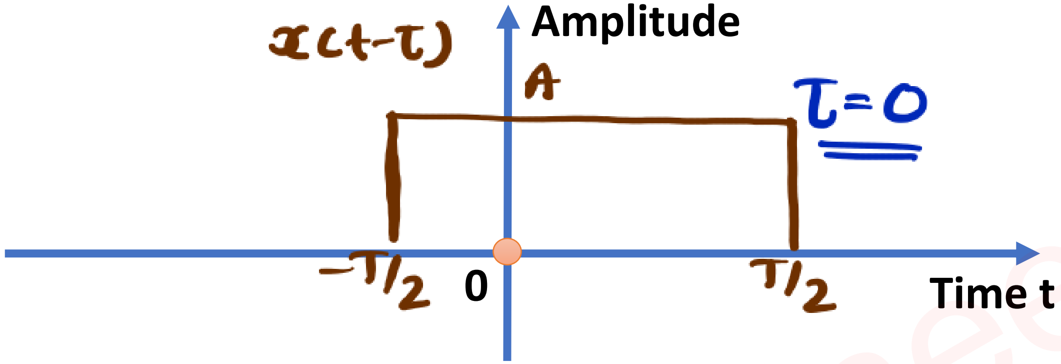
1. Auto Correlation Function
2. Energy of Signal $x(t)$.
3. Auto Correlation of shifted Signal $x(t-2T)$.

$$\rightsquigarrow R_{xx}(\tau) = \int_{-\infty}^{\infty} x(t)\, x(t-\tau)\, dt$$

$$= x(\tau) * x(-\tau)$$

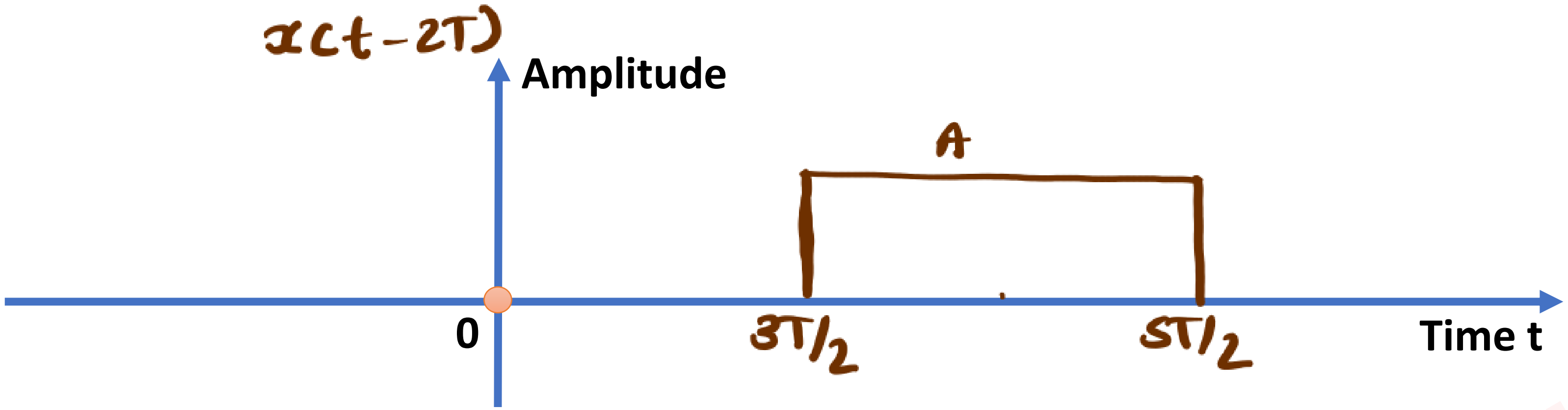




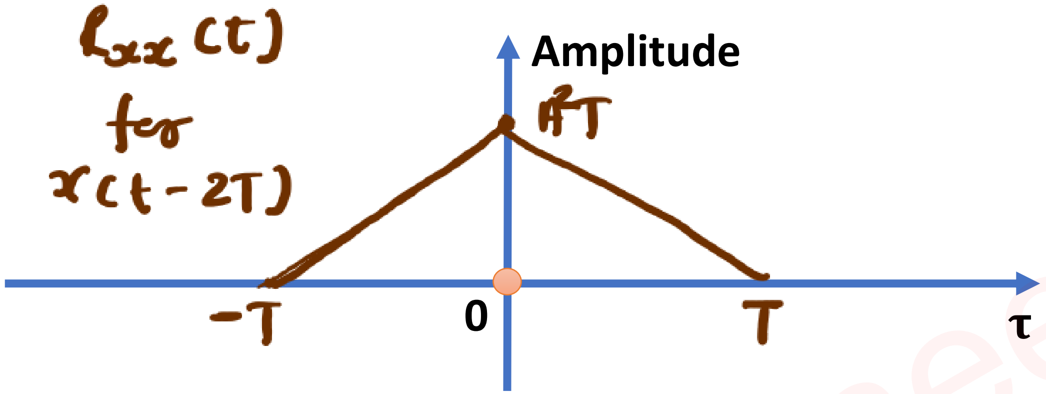


- Lower limits = $-T/2 - T/2 = -T$
- Higher limits = $T/2 + T/2 = T$

$\Rightarrow E = R_{xx}(\tau)\Big|_{\tau=0} = A^2T$



x(t-2T)
Amplitude
A
0
3T/2
5T/2
Time t



Rxx(τ)
for
x(t-2T)
Amplitude
A²T
-T
0
T
τ

Example of Auto Correlation

Example 3. Find the Auto Correlation of $x(t) = e^{-at}u(t)$ and also find Energy of given Signal.

$$\rightsquigarrow R_{xx}(\tau) = \int_{-\infty}^{\infty} x(t)\,x(t-\tau)\,dt$$

$$\rightsquigarrow x(t) = e^{-at}u(t)$$

$$x(t-\tau) = e^{-a(t-\tau)}u(t-\tau)$$

$$\Rightarrow R_{xx}(\tau) = \int_{-\infty}^{\infty} e^{-at}u(t)\,e^{-a(t-\tau)}u(t-\tau)\,dt$$

$$\Rightarrow u(t)u(t-\tau) = 1, \quad t > \tau$$

$$\Rightarrow R_{xx}(\tau) = \int_{\tau}^{\infty} e^{-at}\,e^{-at}\,e^{a\tau}\,dt$$

$$= e^{a\tau}\int_{\tau}^{\infty} e^{-2at}\,dt$$

$$= e^{a\tau}\left[\frac{e^{-2at}}{-2a}\right]_{\tau}^{\infty}$$

$$= \frac{e^{a\tau}}{-2a}\left[\underset{x}{\underline{e^{-\infty}}} - e^{-2a\tau}\right]$$

$$\Rightarrow R_{xx}(\tau) = \frac{1}{2a}e^{-a\tau}$$

- $R_{xx}(\tau)$ is even function, $R_{xx}(t) = R_{xx}(-\tau)$

$$\Rightarrow R_{xx}(t) = \frac{1}{2a}e^{-a|\tau|}$$

- $E = R_{xx}(t)\Big|_{\tau=0} = \frac{1}{2a}$

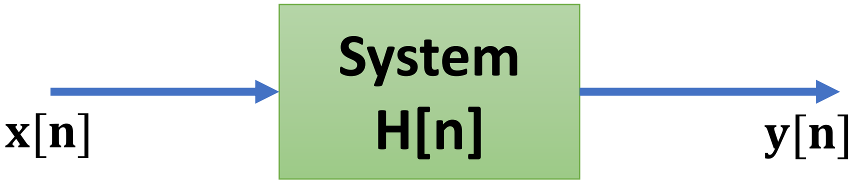
Example of Correlation

Example 1. Write the following terms in form of $R_{xx}[k]$.

a. $R_{xy}[k]$

b. $R_{yx}[k]$

c. $R_{yy}[k]$



$y[n] = x[n] * H[n]$

$$R_{xy}[k] = x[k] * y[-k]$$
$$= x[k] * \{x[-k] * H[-k]\}$$
$$= \{x[k] * x[-k]\} * H[-k]$$
$$= R_{xx}[k] * H[-k]$$

$$R_{yx}[k] = y[k] * x[-k]$$
$$= x[k] * H[k] * x[-k]$$
$$= \{x[k] * x[-k]\} * H[k]$$
$$= R_{xx}[k] * H[k]$$

$$R_{yy}[k] = y[k] * y[-k]$$
$$= x[k] * H[k] * x[-k] * H[-k]$$
$$= \{x[k] * x[-k]\} * \{H[k] * H[-k]\}$$
$$= R_{xx}[k] * R_{HH}[k]$$

Parseval's Theorem for Energy Signals

Statement

- Parseval's theorem states that the total energy of a signal in the time domain is equal to the total energy in the frequency domain.

$$\Rightarrow E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |x(f)|^2 df = \frac{1}{2\pi}\int_{-\infty}^{\infty} |x(\omega)|^2 d\omega$$

Proof

$$\Rightarrow E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} x(t)\, x^*(t)\, dt$$

$$\left[IFT \Rightarrow x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} x(\omega)\, e^{j\omega t}\, d\omega \right.$$

$$\left. \Rightarrow x^*(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} x^*(\omega)\, e^{-j\omega t}\, d\omega \right]$$

$$\Rightarrow E = \int_{-\infty}^{\infty} x(t) \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} x^{*}(\omega)\, e^{-j\omega t}\, d\omega \right) dt$$

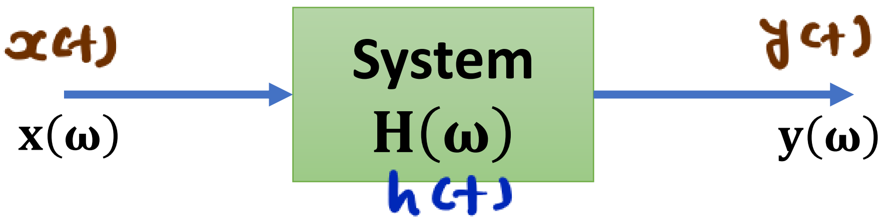
$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} x^{*}(\omega) \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt\, d\omega$$

$$\left[\text{F.T.} \Rightarrow x(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt \right]$$

$$\Rightarrow E = \frac{1}{2\pi} \int_{-\infty}^{\infty} x^{*}(\omega)\, x(\omega)\, d\omega$$

$$\boxed{\Rightarrow E = \frac{1}{2\pi} \int_{-\infty}^{\infty} |x(\omega)|^{2}\, d\omega}$$

Relation of Energy Spectral Density of Input and Output



$\rightarrow$ $y(t) = x(t) * h(t)$

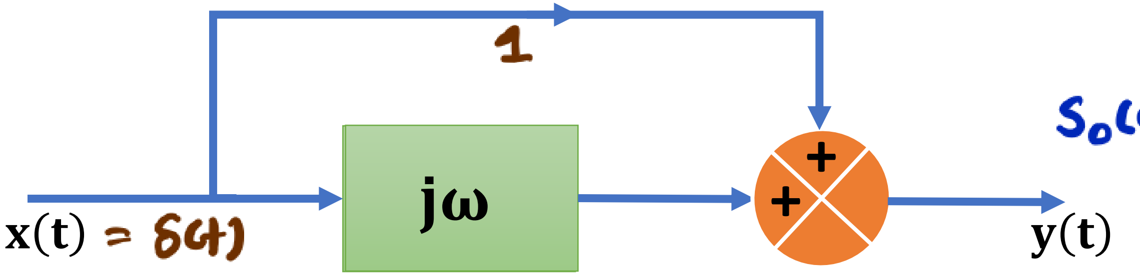
* In freq. domain.

$\Rightarrow$ $Y(\omega) = X(\omega) H(\omega)$

$\Rightarrow$ $|Y(\omega)|^2 = |X(\omega)|^2 |H(\omega)|^2$

$\Rightarrow$ $S_o(\omega) = S_I(\omega) |H(\omega)|^2$

Example Find the output spectral density if input $x(t) = \delta(t)$.



$S_o(\omega) = |Y(\omega)|^2 = ?$

$\rightarrow X(\omega) = 1 \Rightarrow |X(\omega)|^2 = S_I(\omega) = 1$

$\rightarrow \dfrac{Y(\omega)}{X(\omega)} = T.F = 1 + j\omega = H(\omega)$

$$\rightarrow S_o(\omega) = S_I(\omega)\,|H(\omega)|^2$$
$$= 1\,(1+\omega^2)$$
$$= \omega^2 + 1$$